



NSW Education Standards Authority

2019 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks: 100

Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7–17)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1** What is the value of $(3 - 2i)^2$?
- A. $5 - 12i$
B. $5 + 12i$
C. $13 - 12i$
D. $13 + 12i$
- 2** Which of the following is a primitive of $\frac{\sin x}{\cos^3 x}$?
- A. $\frac{1}{2}\sec^2 x$
B. $-\frac{1}{2}\sec^2 x$
C. $\frac{1}{4}\sec^4 x$
D. $-\frac{1}{4}\sec^4 x$
- 3** Which expression is equal to $\int x \cos x \, dx$?
- A. $-x \sin x + \cos x + C$
B. $-x \sin x - \cos x + C$
C. $x \sin x + \cos x + C$
D. $x \sin x - \cos x + C$

- 4 The polynomial $2x^3 + bx^2 + cx + d$ has roots 1 and -3 , with one of them being a double root.

What is a possible value of b ?

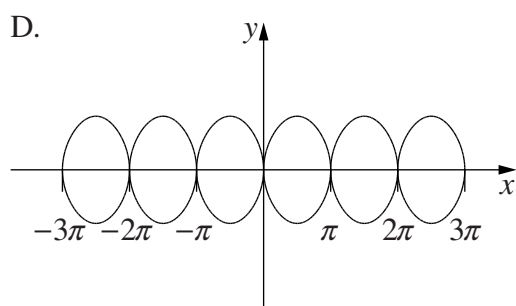
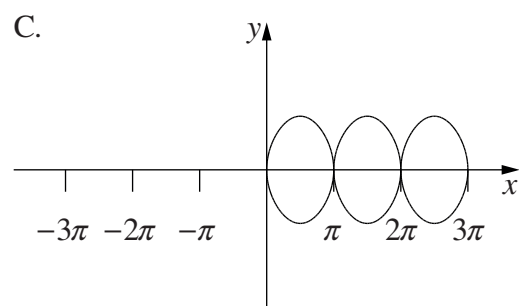
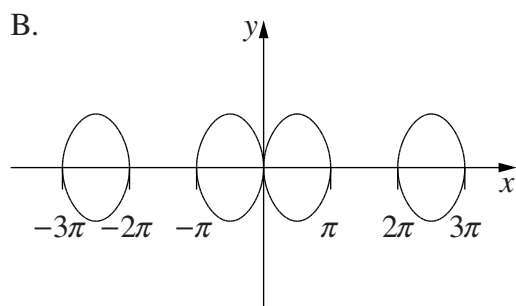
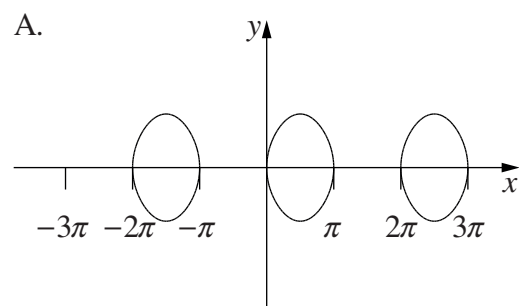
- A. -10
- B. -5
- C. 5
- D. 10

- 5 Which of the following best describes the locus of points in the Argand plane given by

$$|z - 2| + |z + 1| = 7?$$

- A. An ellipse with foci at $(-3, 0)$ and $(4, 0)$
- B. An ellipse with foci at $(-1, 0)$ and $(2, 0)$
- C. A hyperbola with foci at $(-3, 0)$ and $(4, 0)$
- D. A hyperbola with foci at $(-1, 0)$ and $(2, 0)$

- 6 Which graph best represents $y^2 = 2 \sin|x|$?



7 Which of these integrals has the largest value?

A. $\int_0^{\frac{\pi}{4}} \tan x \, dx$

B. $\int_0^{\frac{\pi}{4}} \tan^2 x \, dx$

C. $\int_0^{\frac{\pi}{4}} 1 - \tan x \, dx$

D. $\int_0^{\frac{\pi}{4}} 1 - \tan^2 x \, dx$

8 Let z be a complex number such that $z^2 = -i\bar{z}$.

Which of the following is a possible value for z ?

A. $\frac{1}{2} - \frac{\sqrt{3}}{2}i$

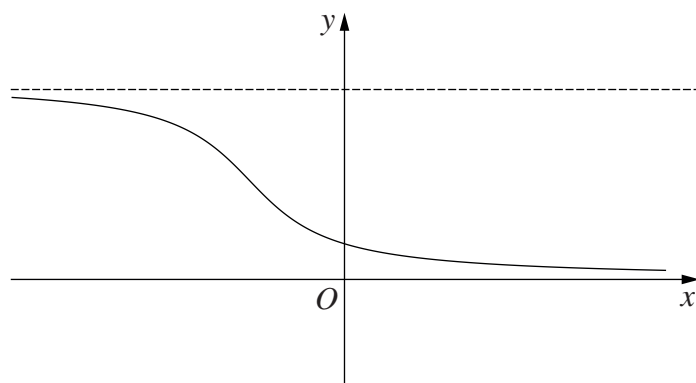
B. $\frac{1}{2} + \frac{\sqrt{3}}{2}i$

C. $\frac{\sqrt{3}}{2} - \frac{1}{2}i$

D. $\frac{\sqrt{3}}{2} + \frac{1}{2}i$

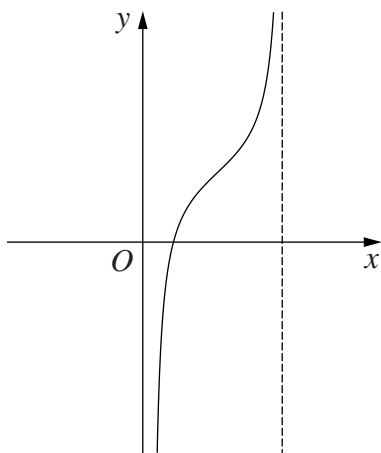
- 9 It is given that the function $f(x)$ has an inverse. Let $g(x) = f^{-1}(x)$.

The diagram shows the graph of the derivative $y = f'(x)$.

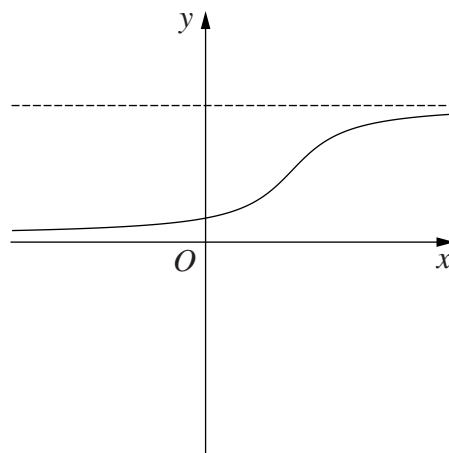


Which graph best represents the graph of $y = g'(x)$?

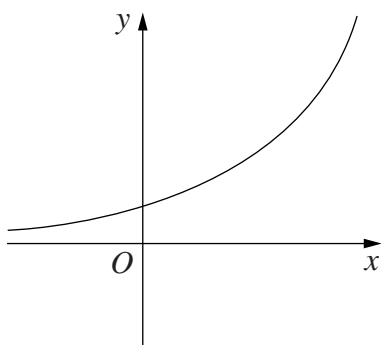
A.



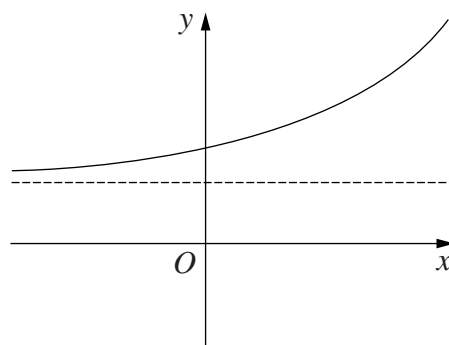
B.



C.



D.



- 10** An access code consists of 4 digits chosen from the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. The code will only work if the digits are entered in the correct order.

Some access codes contain exactly two different digits, for example 3377 or 5155.

How many such access codes can be made using exactly two different digits?

- A. 630
- B. 900
- C. 1080
- D. 2160

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

(a) Let $z = 1 + 3i$ and $w = 2 - i$.

(i) Find $z + \bar{w}$. **1**

(ii) Express $\frac{z}{w}$ in the form $x + iy$, where x and y are real numbers. **2**

(b) Sketch the ellipse $\frac{x^2}{4} + y^2 = 1$, showing the intercepts, foci and directrices. **3**

(c) Find $\int \frac{dx}{x^2 + 10x + 29}$. **2**

(d) Find $\int \frac{6}{x^2 - 9} dx$. **3**

(e) Let $z = -1 + i\sqrt{3}$.

(i) Write z in modulus–argument form. **2**

(ii) Find z^3 , giving your answer in the form $x + iy$, where x and y are real numbers. **2**

Question 12 (15 marks) Use the Question 12 Writing Booklet.

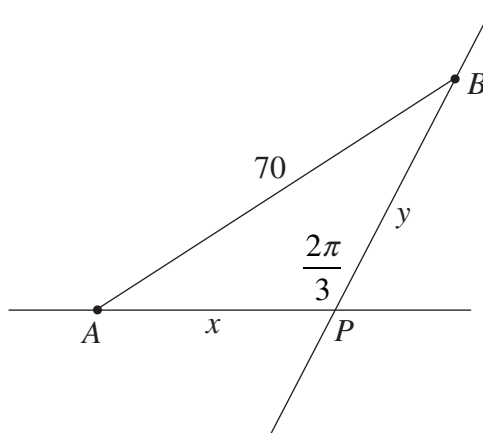
- (a) Sketch the region defined by $\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{2}$ and $\operatorname{Im}(z) \leq 1$.

2

- (b) The diagram shows two straight railway tracks that meet at an angle of $\frac{2\pi}{3}$ at the point P .

Trains A and B are joined by a cable which is 70 m long.

At time t seconds, train A is x metres from P and train B is y metres from P .



Train B is towing train A and is moving at a constant speed of 4 m s^{-1} away from P .

- (i) Show that $x^2 + xy + y^2 = 70^2$.

1

- (ii) What is the value of $\frac{dx}{dt}$ when train A is 30 metres from P and train B is 50 metres from P ?

3

Question 12 continues on page 9

Question 12 (continued)

- (c) Following a volcanic eruption, fine ash is deposited in a circular pattern around the point of eruption. **4**

The depth in kilometres, $D(x)$, of the fine ash can be modelled by

$$D(x) = Ae^{-kx},$$

where x is the distance in kilometres from the point of eruption and $k > 0$.

Using the method of cylindrical shells, find an expression for the volume of fine ash deposited between 10 km and 40 km from the point of eruption.

- (d) Consider the function $f(x) = x^3 - 1$.

(i) Sketch the graph $y = |f(x)|$. **1**

(ii) Sketch the graph $y = \frac{1}{f(x)}$. **2**

(iii) Without using calculus, sketch the graph $y = \frac{x}{f(x)}$. **2**

End of Question 12

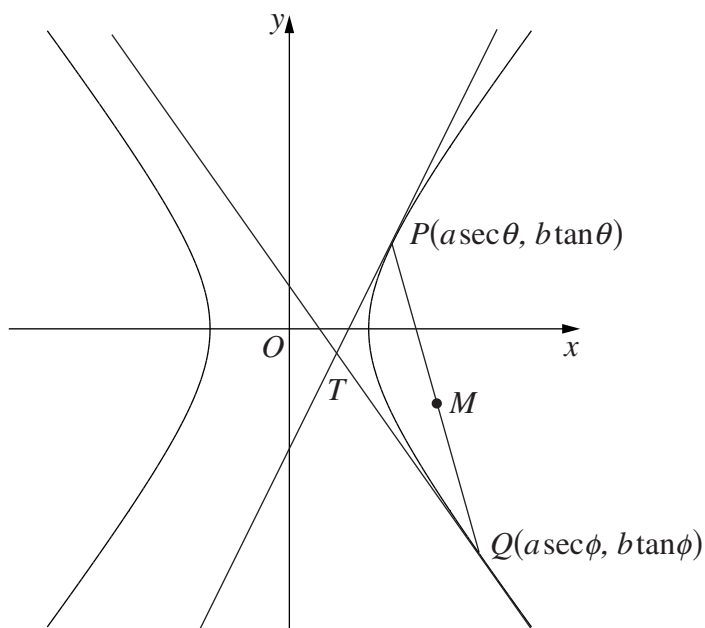
Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) Two points $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \phi, b \tan \phi)$ are on the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Let O be the origin and let M be the midpoint of PQ .

Tangents to the hyperbola are drawn at points P and Q as shown.



- (i) Show that the equation of the tangent at P is 2

$$bx \sec \theta - ay \tan \theta = ab.$$

- (ii) The tangents at P and Q meet at the point $T(x_0, y_0)$. 2

Show that $x_0 = \frac{a(\tan \theta - \tan \phi)}{\sec \phi \tan \theta - \sec \theta \tan \phi}.$

- (iii) It is given that $y_0 = \frac{b(\sec \theta - \sec \phi)}{\sec \phi \tan \theta - \sec \theta \tan \phi}$. (Do NOT prove this.) 3

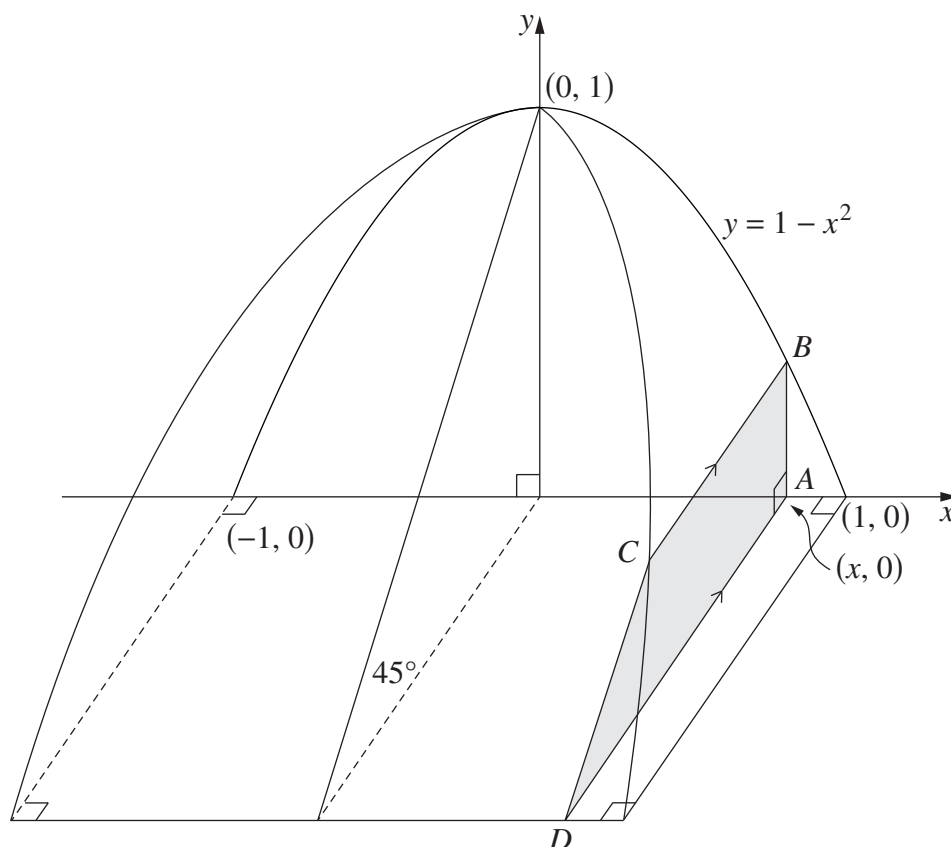
By considering gradients, or otherwise, show that O , T and M are collinear.

Question 13 continues on page 11

Question 13 (continued)

- (b) One end of a solid is the region bounded by $y = 1 - x^2$ and the x -axis. The base of the solid is a rectangle perpendicular to the x, y -plane with one side lying on the x -axis. A flat face of the solid is formed by the plane passing through the point $(0, 1)$ making an angle of 45° to the base, as shown in the diagram.

Vertical cross-sections perpendicular to the x -axis are trapezia. One trapezium $ABCD$ is shown, where A is the point $(x, 0)$.



- (i) Show that the area of $ABCD$ is $\frac{1 - x^4}{2}$. 2
- (ii) Find the volume of the solid. 1

Question 13 continues on page 12

Question 13 (continued)

- (c) Two objects are projected from the same point on a horizontal surface. Object 1 is projected with an initial velocity of 20 m s^{-1} directed at an angle of $\frac{\pi}{3}$ to the horizontal. Object 2 is projected 2 seconds later.

The equations of motion of an object projected from the origin with initial velocity v at an angle θ to the x -axis are

$$x = vt \cos \theta$$
$$y = -4.9t^2 + vt \sin \theta,$$

where t is the time after the projection of the object. Do NOT prove these equations.

- (i) Show that Object 1 will land at a distance $\frac{100\sqrt{3}}{4.9}$ m from the point of projection. **2**
- (ii) The two objects hit the horizontal plane at the same place and time. **3**

Find the initial speed and the angle of projection of Object 2, giving your answer correct to 1 decimal place.

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) A cyclist rides around a track. The track is a circle of radius r metres, banked at an angle of θ to the horizontal. **3**

Let $v \text{ m s}^{-1}$ be the velocity at which the cyclist experiences no net force parallel to the track.

By resolving forces, show that v satisfies $v^2 = gr \tan \theta$ where g is the acceleration due to gravity.

- (b) A parachutist jumps from a plane, falls freely for a short time and then opens the parachute. Let t be the time in seconds after the parachute opens, $x(t)$ be the distance in metres travelled after the parachute opens, and $v(t)$ be the velocity of the parachutist in m s^{-1} .

The acceleration of the parachutist after the parachute opens is given by

$$\ddot{x} = g - kv,$$

where $g \text{ m s}^{-2}$ is the acceleration due to gravity and k is a positive constant.

- (i) With an open parachute the parachutist has a terminal velocity of $w \text{ m s}^{-1}$. **1**

Show that $w = \frac{g}{k}$.

At the time the parachute opens, the speed of descent is $1.6w \text{ m s}^{-1}$.

- (ii) Show that it takes $\frac{1}{k} \log_e 6$ seconds to slow down to a speed of $1.1w \text{ m s}^{-1}$. **4**

- (iii) Let D be the distance the parachutist travels between opening the parachute and reaching the speed $1.1w \text{ m s}^{-1}$. **3**

Show that $D = \frac{g}{k^2} \left(\frac{1}{2} + \log_e 6 \right)$.

- (c) (i) Show that $\cot x - \cot 2x = \operatorname{cosec} 2x$. **2**

- (ii) Use mathematical induction to prove that, for all $n \geq 1$, **2**

$$\sum_{r=1}^n \operatorname{cosec}(2^r x) = \cot x - \cot(2^n x).$$

Question 15 (15 marks) Use the Question 15 Writing Booklet.

(a) (i) Show that $\int_{-a}^a \frac{f(x)}{f(x) + f(-x)} dx = \int_{-a}^a \frac{f(-x)}{f(x) + f(-x)} dx.$ **2**

(ii) Hence, or otherwise, evaluate $\int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx.$ **2**

- (b) An urn contains w white marbles and y yellow marbles. Two players, A and B , play a game in which they take turns to randomly draw a marble from the urn.

If player A draws a white marble then player A wins. If player B draws a yellow marble then player B wins.

Each player replaces the marble if their draw is unsuccessful. The game stops as soon as one player wins.

Player A draws first.

- (i) Explain why, in the case where $w = y$, player A has a greater chance of winning than player B . **1**

- (ii) Show that for player B to have a greater chance of winning than player A , **3**

$$\frac{y}{w} > \frac{1 + \sqrt{5}}{2}.$$

(c) (i) Show that $\int_0^1 \frac{x}{(x+1)^2} dx = \ln 2 - \frac{1}{2}.$ **2**

(ii) Let $I_n = \int_0^1 \frac{x^n}{(x+1)^2} dx.$ **3**

Show that $I_n = \frac{1}{2(n-1)} - \frac{n}{n-1} I_{n-1}$ for $n \geq 2.$

- (iii) Evaluate $I_3.$ **2**

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) (i) Consider the equation $x^3 - px + q = 0$, where p and q are real numbers and $p > 0$. **2**

Let $r = \sqrt{\frac{4p}{3}}$ and $\cos 3\theta = \frac{-4q}{r^3}$.

Show that $r \cos \theta$ is a root of $x^3 - px + q = 0$.

You may use the result $4 \cos^3 \theta - 3 \cos \theta = \cos 3\theta$. (Do NOT prove this.)

- (ii) Let α, β and γ be the roots of the equation **2**

$$x^3 + 9x^2 + 15x - 17 = 0.$$

Show that $\alpha + 3, \beta + 3, \gamma + 3$ are the roots of $x^3 - 12x - 8 = 0$.

- (iii) Using parts (i) and (ii), solve **3**

$$x^3 + 9x^2 + 15x - 17 = 0.$$

Question 16 continues on page 16

Question 16 (continued)

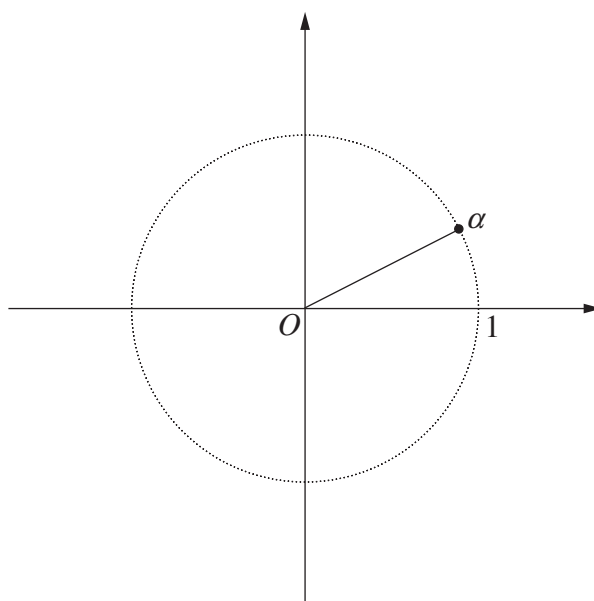
(b) Let $P(z) = z^4 - 2kz^3 + 2k^2z^2 + mz + 1$, where k and m are real numbers.

The roots of $P(z)$ are $\alpha, \bar{\alpha}, \beta, \bar{\beta}$.

It is given that $|\alpha| = 1$ and $|\beta| = 1$.

(i) Show that $(\operatorname{Re}(\alpha))^2 + (\operatorname{Re}(\beta))^2 = 1$. **3**

(ii) The diagram shows the position of α . **2**



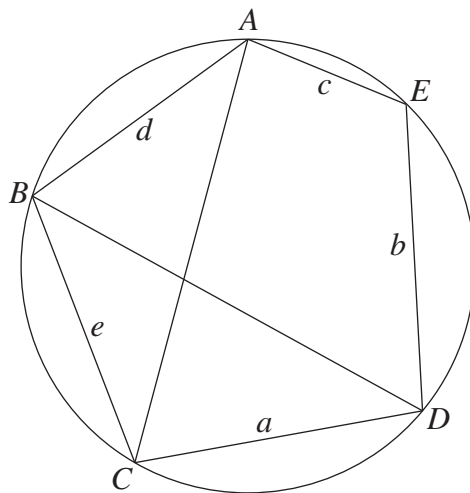
Copy or trace the diagram into your writing booklet.

On the diagram, accurately show all possible positions of β .

Question 16 continues on page 17

Question 16 (continued)

- (c) The vertices A, B, C, D and E of a pentagon lie on a circle. The lengths of the sides opposite A, B, C, D and E are a, b, c, d and e , respectively, as shown. **3**



Copy or trace the diagram into your writing booklet.

Using the same convention as in the sine rule, the vertex name also represents the size of the angle at that vertex. For example, A is used to represent the size of $\angle BAE$.

Show that $\frac{a}{\sin(B + E)} = \frac{d}{\sin(C + E)}$.

End of paper

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NSW Education Standards Authority

2019 HIGHER SCHOOL CERTIFICATE EXAMINATION

REFERENCE SHEET

- Mathematics –
- Mathematics Extension 1 –
- Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

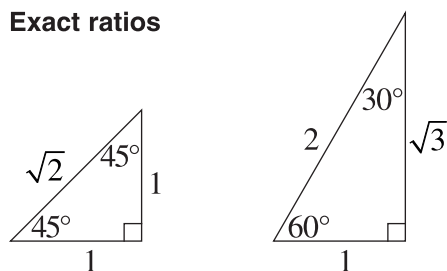
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

$$\text{tangent: } y = tx - at^2$$

$$\text{normal: } x + ty = at^3 + 2at$$

At (x_1, y_1) ,

$$\text{tangent: } xx_1 = 2a(y + y_1)$$

$$\text{normal: } y - y_1 = -\frac{2a}{x_1}(x - x_1)$$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2019 HSC Mathematics Extension 2 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	A
2	A
3	C
4	D
5	B
6	B
7	D
8	C
9	D
10	A

Section II

Question 11 (a) (i)

Criteria	Marks
Provides correct solution	1

Sample answer:

$$\begin{aligned}
 & z + \bar{w} \\
 &= 1 + 3i + 2 + i \\
 &= 3 + 4i
 \end{aligned}$$

Question 11 (a) (ii)

Criteria	Marks
Provides correct solution	2
Multiplies numerator and denominator by the conjugate of w, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \frac{z}{w} \\
 &= \frac{1+3i}{2-i} \times \frac{2+i}{2+i} \\
 &= \frac{(1+3i)(2+i)}{(2-i)(2+i)} \\
 &= \frac{2+6i+i-3}{4-(-1)} \\
 &= \frac{7i-1}{5} \\
 &= \frac{-1}{5} + \frac{7i}{5}
 \end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	3
• Determines the location of the foci and the equations of the directrices, or equivalent merit	2
• Sketches an ellipse, showing correct intercepts, or equivalent merit	1

Sample answer:

$$\frac{x^2}{4} + y^2 = 1$$

$$a = 2$$

$$b = 1$$

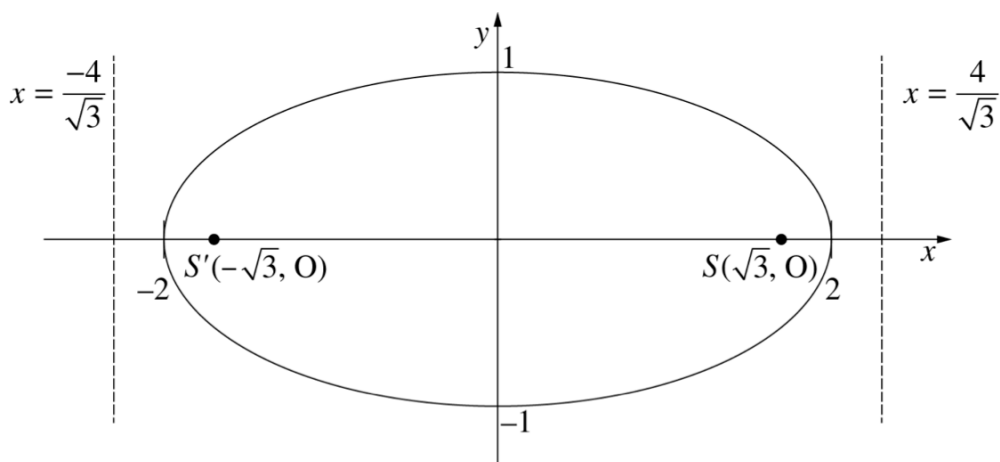
$$b^2 = a^2(1 - e^2) \quad \text{Foci at } x = \pm ae = \pm\sqrt{3}$$

$$1 = 4(1 - e^2) \quad \text{directrices } x = \pm \frac{a}{e} = \pm \frac{4}{\sqrt{3}}$$

$$1 - e^2 = \frac{1}{4}$$

$$e^2 = \frac{3}{4}$$

$$e = \frac{\sqrt{3}}{2}$$



Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Completes the square of the denominator, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \int \frac{dx}{x^2 + 10x + 29} \\
 &= \int \frac{dx}{(x+5)^2 + 4} \\
 &= \int \frac{dx}{(x+5)^2 + 2^2} \\
 &= \frac{1}{2} \tan^{-1} \left(\frac{x+5}{2} \right) + c
 \end{aligned}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	3
• Correctly writes the integrand as the sum of partial fractions, or equivalent merit	2
• Attempts to write the integrand as the sum of partial fractions, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \int \frac{6}{x^2 - 9} dx \\
 & \frac{6}{x^2 - 9} = \frac{a}{x+3} + \frac{b}{x-3} \\
 & a(x-3) + b(x+3) = 6 \\
 & \text{Let } x = 3 \qquad \qquad \text{and } x = -3 \\
 & b(6) = 6 \qquad \qquad a(-6) = 6 \\
 & \therefore b = 1 \qquad \qquad a = -1 \\
 & \therefore \int \frac{1}{x-3} - \frac{1}{x+3} dx \\
 &= \ln |x-3| - \ln |x+3| + c
 \end{aligned}$$

Question 11 (e) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains the modulus, or equivalent merit	1

Sample answer:

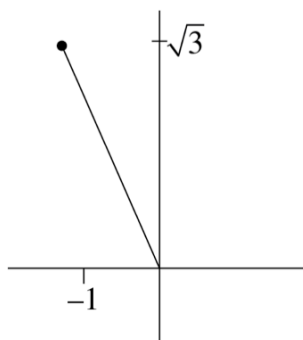
$$z = -1 + i\sqrt{3}$$

$$r = \sqrt{1+3} \quad \tan \theta = \frac{\sqrt{3}}{1}$$

$$r = \sqrt{4} \quad \theta = \frac{\pi}{3}$$

$$= 2 \quad \arg(z) = \frac{2\pi}{3}$$

$$z = 2 \left(\cos \left(\frac{2\pi}{3} \right) + i \sin \left(\frac{2\pi}{3} \right) \right)$$



Question 11 (e) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses de Moivre's theorem, or equivalent merit	1

Sample answer:

$$z^3 = 2^3 \left(\cos 3 \left(\frac{2\pi}{3} \right) + i \sin 3 \left(\frac{2\pi}{3} \right) \right)$$

$$= 8(\cos 2\pi + i \sin 2\pi)$$

$$= 8(1 + 0i)$$

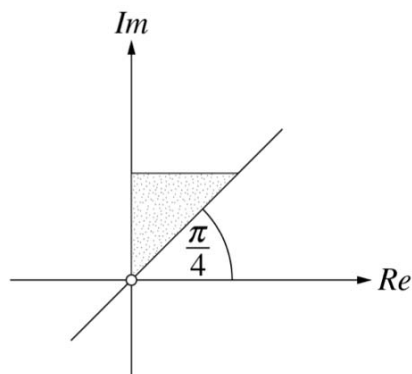
$$= 8 + 0i$$

$$\therefore x = 8, y = 0$$

Question 12 (a)

Criteria	Marks
• Provides correct solution	2
• Correctly interprets one condition, or equivalent merit	1

Sample answer:



Question 12 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$x^2 + y^2 - 2xy \cos \frac{2\pi}{3} = 70^2$$

$$x^2 + y^2 - 2xy \left(\frac{-1}{2} \right) = 70^2$$

$$x^2 + y^2 + xy = 70^2$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds the value of $\frac{dy}{dx}$, or equivalent merit	2
• Implicitly differentiates $x^2 + xy + y^2 = 70^2$ with respect to x , or equivalent merit	1

Sample answer:

$$x^2 + xy + y^2 = 70^2$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(xy) + \frac{d}{dx}(y^2) = \frac{d}{dx}(70^2)$$

$$2x + x \cdot \frac{dy}{dx} + y \cdot 1 + 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-(y + 2x)}{x + 2y}$$

$$\text{at } x = 30 \ y = 50$$

$$\frac{dy}{dx} = \frac{-(50 + 2(30))}{30 + 2(50)}$$

$$= \frac{-110}{130}$$

$$= \frac{-11}{13}$$

$$\frac{dx}{dt} = \frac{dx}{dy} \cdot \frac{dy}{dt}$$

$$\therefore \frac{dx}{dt} = \frac{-13}{11} \times 4$$

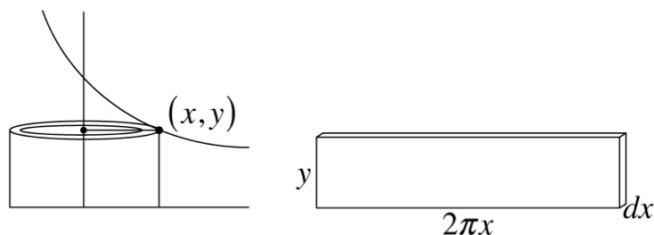
$$= \frac{-52}{11}$$

$$= -4.7$$

Question 12 (c)

Criteria	Marks
• Provides correct solution	4
• Correctly applies integration by parts to an integral for the volume, or equivalent merit	3
• Obtains an integral from which the volume can be evaluated, or equivalent merit	2
• Obtains the volume of a shell, or equivalent merit	1

Sample answer:

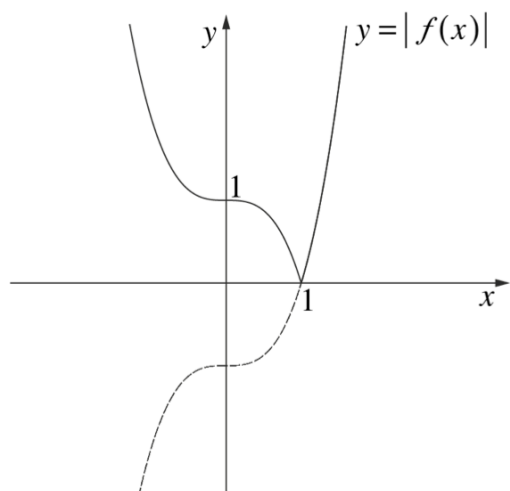


$$\begin{aligned}
 \text{Volume} &= 2A\pi \int_{10}^{40} x e^{-kx} dx \\
 &= 2A\pi \left[\left[-\frac{1}{k} x e^{-kx} \right]_{10}^{40} + \frac{1}{k} \int_{10}^{40} e^{-kx} dx \right] \\
 &= 2A\pi \left[\left[-\frac{1}{k} x e^{-kx} \right]_{10}^{40} + \frac{1}{k} \left[\frac{e^{-kx}}{-k} \right]_{10}^{40} \right] \\
 &= 2A\pi \left[\left(\frac{-40}{k} e^{-40k} + \frac{10}{k} e^{-10k} \right) - \frac{1}{k^2} (e^{-40k} - e^{-10k}) \right]
 \end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
Provides correct sketch	1

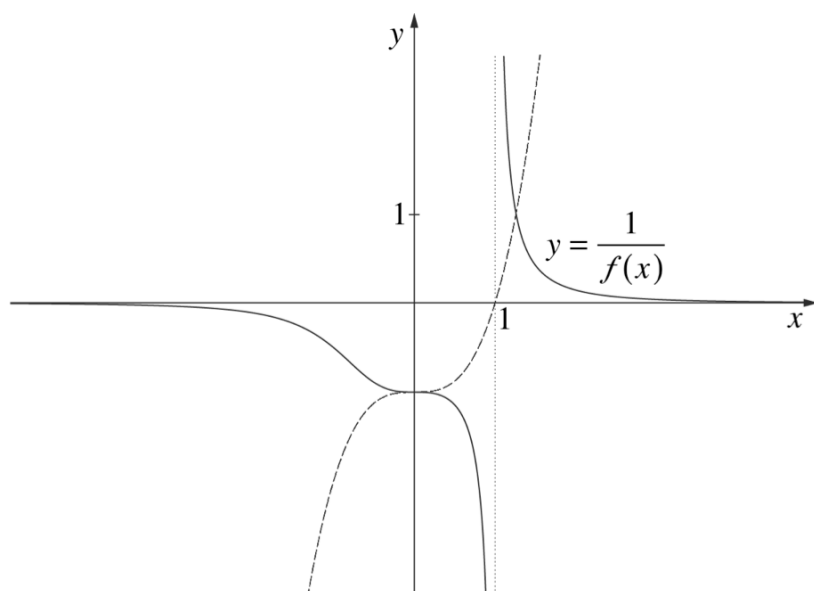
Sample answer:



Question 12 (d) (ii)

Criteria	Marks
Provides correct sketch	2
Provides sketch with correct asymptote, or equivalent merit	1

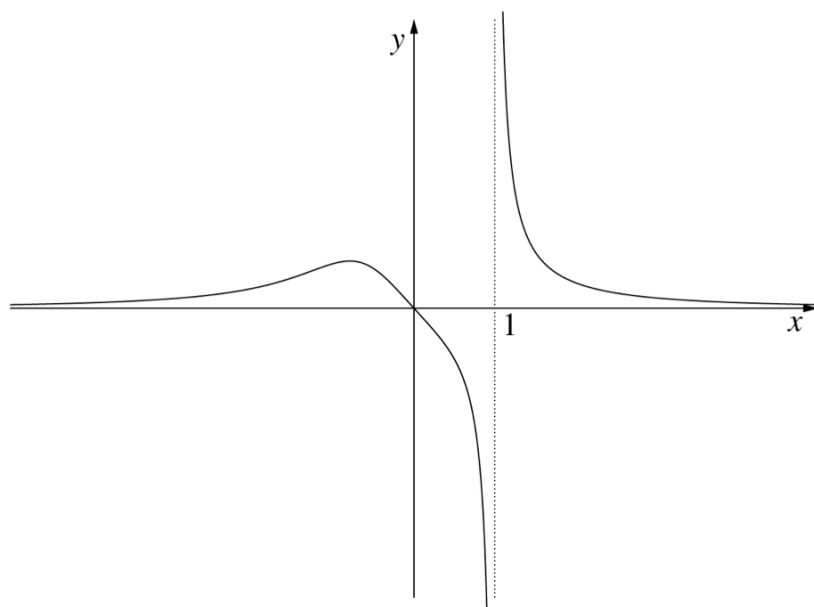
Sample answer:



Question 12 (d) (iii)

Criteria	Marks
• Provides correct sketch	2
• Provides sketch of curve passing through the origin which is not defined at $x = 1$, or equivalent merit	1

Sample answer:



Question 13 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Finds the slope of the hyperbola at P , or equivalent merit	1

Sample answer:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{d}{dx} \left(\frac{x^2}{a^2} \right) - \frac{d}{dx} \left(\frac{y^2}{b^2} \right) = \frac{d}{dx} (1)$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \left(\frac{dy}{dx} \right) = 0$$

$$\frac{2x}{a^2} = \frac{2y}{b^2} \left(\frac{dy}{dx} \right)$$

$$\frac{xb^2}{ya^2} = \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{xb^2}{ya^2}$$

at $P(a \sec \theta, b \tan \theta)$

$$m = \frac{b^2 a \sec \theta}{a^2 b \tan \theta}$$

$$m = \frac{b \sec \theta}{a \tan \theta}$$

Equation of tangent at P .

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$a \tan \theta (y - b \tan \theta) = b \sec \theta (x - a \sec \theta)$$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$ab \sec^2 \theta - ab \tan^2 \theta = bx \sec \theta - ay \tan \theta$$

$$ab (\sec^2 \theta - \tan^2 \theta) = bx \sec \theta - ay \tan \theta$$

$$\therefore ab = bx \sec \theta - ay \tan \theta$$

Question 13 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to find the point of intersection of the two tangents, or equivalent merit	1

Sample answer:

$$bx \sec \theta = ay \tan \theta + ab \quad (\text{tangent at } P)$$

$$\therefore \frac{bx \sec \theta - ab}{a \tan \theta} = y$$

$$bx \sec \phi = ay \tan \phi + ab \quad (\text{tangent at } Q)$$

$$\therefore \frac{bx \sec \phi - ab}{a \tan \phi} = y$$

$$\therefore \frac{bx \sec \theta - ab}{a \tan \theta} = \frac{bx \sec \phi - ab}{a \tan \phi}$$

$$a \tan \phi (bx \sec \theta - ab) = a \tan \theta (bx \sec \phi - ab)$$

$$abx \sec \theta \tan \phi - a^2 b \tan \phi = abx \tan \theta \sec \phi - a^2 b \tan \theta$$

$$a^2 b \tan \theta - a^2 b \tan \phi = abx \tan \theta \sec \phi - abx \sec \theta \tan \phi$$

$$a^2 b (\tan \theta - \tan \phi) = abx (\tan \theta \sec \phi - \sec \theta \tan \phi)$$

At the point of intersection

$$x_0 = \frac{a(\tan \theta - \tan \phi)}{\tan \theta \sec \phi - \sec \theta \tan \phi}$$

Question 13 (a) (iii)

Criteria	Marks
• Provides correct solution	3
• Finds the slopes of OM and OT , or equivalent merit	2
• Finds the slope of OM , or equivalent merit	1

Sample answer:

Find M .

$$M = \left(\frac{a(\sec \theta + \sec \phi)}{2}, \frac{b(\tan \theta + \tan \phi)}{2} \right)$$

Find gradient of OM .

$$\frac{\frac{b(\tan \theta + \tan \phi)}{2}}{\frac{a(\sec \theta + \sec \phi)}{2}} = \frac{b(\tan \theta + \tan \phi)}{a(\sec \theta + \sec \phi)}$$

Find gradient of OT .

$$\frac{\frac{b(\sec \theta - \sec \phi)}{\sec \phi \tan \theta - \sec \theta \tan \phi}}{\frac{a(\tan \theta - \tan \phi)}{\sec \phi \tan \theta - \sec \theta \tan \phi}} = \frac{b(\sec \theta - \sec \phi)}{a(\tan \theta - \tan \phi)}$$

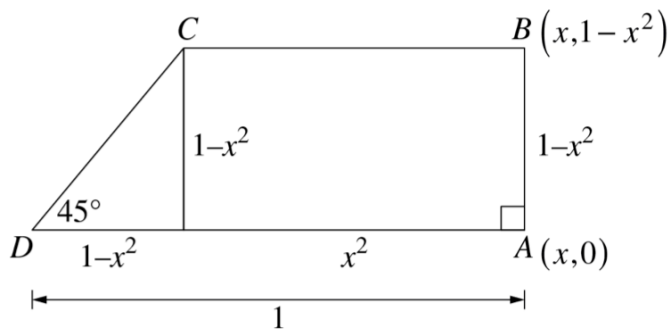
Points will be collinear if the 2 gradients are equal.

$$\begin{aligned} \text{Gradient of } OM &= \frac{b(\tan \theta + \tan \phi)}{a(\sec \theta + \sec \phi)} \times \frac{(\sec \theta - \sec \phi)}{(\sec \theta - \sec \phi)} \\ &= \frac{b(\tan \theta + \tan \phi)(\sec \theta - \sec \phi)}{a(\sec^2 \theta - \sec^2 \phi)} \\ &= \frac{b(\tan \theta + \tan \phi)(\sec \theta - \sec \phi)}{a(1 + \tan^2 \theta - 1 - \tan^2 \phi)} \\ &= \frac{b(\tan \theta + \tan \phi)(\sec \theta - \sec \phi)}{a(\tan^2 \theta - \tan^2 \phi)} \\ &= \frac{b(\cancel{\tan \theta + \tan \phi})(\sec \theta - \sec \phi)}{a(\tan \theta - \tan \phi)(\cancel{\tan \theta + \tan \phi})} \\ &= \frac{b(\sec \theta - \sec \phi)}{a(\tan \theta - \tan \phi)} = \text{gradient of } OT. \end{aligned}$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Finds the length of CB , or equivalent merit	1

Sample answer:



B lies on the parabola $y = 1 - x^2$ so $|AB| = 1 - x^2$.

D lies on line parallel to x -axis and 1 unit away, known from isosceles triangle of height 1 in the centre of the diagram. So $|AD| = 1$.

$$\begin{aligned}
 \text{Area of trapezium } ABCD &= \frac{1}{2}(1 - x^2)[1 + 1 - (1 - x^2)] \\
 &= \frac{1}{2}(1 - x^2)(1 + x^2) \\
 &= \frac{1}{2}(1 - x^4)
 \end{aligned}$$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct volume	1

Sample answer:

$$\begin{aligned}
 V &= \int_{-1}^1 \frac{1}{2}(1-x^4) dx \\
 &= 2 \int_0^1 \frac{1}{2}(1-x^4) dx \\
 &= \int_0^1 (1-x^4) dx \\
 &= \left[x - \frac{x^5}{5} \right]_0^1 = 1 - \frac{1}{5} = \frac{4}{5}
 \end{aligned}$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Finds the time at which Object 1 lands, or equivalent merit	1

Sample answer:

For object 1

$$\begin{aligned}
 y &= -4.9t^2 + 20\sin\frac{\pi}{3}t \\
 &= -4.9t^2 + 10\sqrt{3}t
 \end{aligned}$$

$$\text{When } y = 0 \quad t(-4.9t + 10\sqrt{3}) = 0$$

$$\therefore t = 0 \quad \text{or} \quad t = \frac{10\sqrt{3}}{4.9}$$

$$\text{When } t = \frac{10\sqrt{3}}{4.9}$$

$$\begin{aligned}
 x &= 20 \left(\frac{10\sqrt{3}}{4.9} \right) \cos\frac{\pi}{3} \\
 &= 200 \left(\frac{\sqrt{3}}{4.9} \right) \frac{1}{2} \\
 &= \frac{100\sqrt{3}}{4.9}
 \end{aligned}$$

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Obtains two equations for the speed and angle of projection of Object 2, or equivalent merit	2
• Finds the time of flight of Object 2, or equivalent merit	1

Sample answer:

Let V be the initial speed and θ the angle of projection of object 2. Let T be time (in seconds) after projection.

$$y = -4.9T^2 + VT \sin \theta$$

When $y = 0$ $T(-4.9T + V \sin \theta) = 0$

$$\therefore T = 0 \text{ or } T = \frac{V \sin \theta}{4.9}$$

$$\begin{aligned} \text{Distance travelled} &= V \left(\frac{V \sin \theta}{4.9} \right) \cos \theta \\ &= \frac{V^2 \sin \theta \cos \theta}{4.9} \end{aligned}$$

Equating these distances

$$\frac{V^2 \sin \theta \cos \theta}{4.9} = \frac{100\sqrt{3}}{4.9}$$

$$V^2 \sin \theta \cos \theta = 100\sqrt{3} \quad \text{①}$$

Equating the times

$$\frac{V \sin \theta}{4.9} = \frac{10\sqrt{3}}{4.9} - 2$$

$$V \sin \theta = 10\sqrt{3} - 9.8 \quad \text{②}$$

$$V^2 \sin \theta \cos \theta = 100\sqrt{3} \quad \text{①}$$

$$V \sin \theta = 10\sqrt{3} - 9.8 \approx 7.52$$

$$V^2 \sin^2 \theta = 56.55 \quad \text{③}$$

$$\tan \theta = \frac{56.55}{100\sqrt{3}} \quad \text{③} \div \text{①}$$

$$= 0.326$$

$$\theta = 18.1^\circ$$

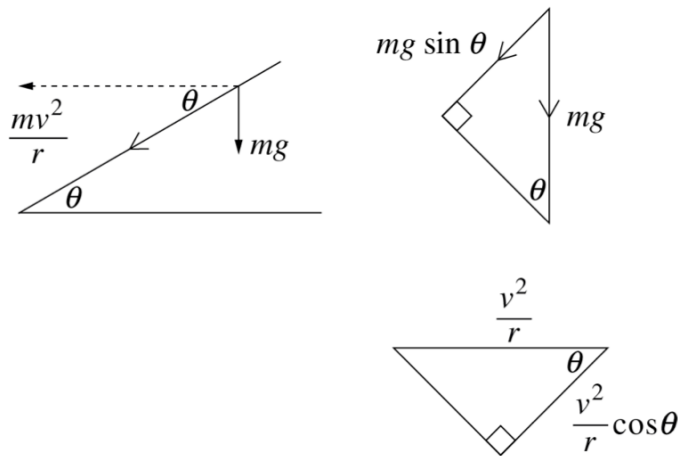
$$V = \frac{7.52}{\sin 18.08^\circ}$$

$$= 24.2$$

Question 14 (a)

Criteria	Marks
• Provides correct solution	3
• Uses the fact that there is no net force parallel to the track to equate forces, or equivalent merit	2
• Correctly resolves forces horizontally and vertically, or equivalent merit	1

Sample answer:



Friction force parallel to slope = mass x acceleration along slope.

$$F + mg \sin \theta = \frac{mv^2}{r} \cos \theta$$

If there are no net forces parallel to track, $F = 0$.

That is,

$$mg \sin \theta = \frac{mv^2}{r} \cos \theta$$

$$\frac{v^2}{r} \cos \theta = g \sin \theta$$

$$\begin{aligned} v^2 &= \frac{gr \sin \theta}{\cos \theta} \\ &= gr \tan \theta \end{aligned}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Terminal velocity is a solution of $\ddot{x} = 0$.

So $0 = g - kv$

$$g = kv$$

$$v = \frac{g}{k}$$

So $w = \text{terminal velocity} = \frac{g}{k}$.

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	4
• Finds the integral and evaluates the constant of integration, or equivalent merit	3
• Finds the integral, or equivalent merit	2
• Obtains $\int \frac{dv}{g - kv} = \int dt$, or equivalent merit	1

Sample answer:

$v = 1.6 w$ when parachute opens. Let that time be $t = 0$.

$v = 1.6 \times \frac{g}{k}$ from previous part.

$$\ddot{x} = \frac{dv}{dt} = g - kv$$

$$\int \frac{dv}{g - kv} = \int dt$$

$$\frac{-1}{k} \ln|g - kv| = t + c$$

$$\text{when } t = 0 \quad v = \frac{1.6g}{k}$$

$$\text{so } \frac{-1}{k} \ln \left| g - k \cdot \frac{1.6g}{k} \right| = c$$

$$\frac{-1}{k} \ln|0.6g| = c$$

$$\text{so } \frac{-1}{k} \ln|g - kv| = t - \frac{1}{k} \ln| -0.6g|$$

Find t when $v = 1.1 w$.

$$\frac{-1}{k} \ln \left| g - k \times 1.1 \times \frac{g}{k} \right| = t - \frac{1}{k} \ln| -0.6g|$$

$$t = \frac{1}{k} \ln| -0.6g| - \frac{1}{k} \ln|g - 1.1g|$$

$$= \frac{1}{k} \ln \left| \frac{-0.6g}{-0.1g} \right| = \frac{1}{k} \ln 6$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	3
• Finds an expression for x in terms of t , or equivalent merit	2
• Obtains an expression for v as a function of t , or equivalent merit	1

Sample answer:

$$\frac{-1}{k} \ln |g - kv| = t - \frac{1}{k} \ln |-0.6g|$$

$$\ln |g - kv| = -kt + \ln |-0.6g|$$

$$|g - kv| = e^{-kt + \ln |-0.6g|}$$

$$= |-0.6g| e^{-kt}$$

$$g - kv = -0.6g e^{-kt}, \text{ since } g - kv < 0$$

$$-kv = -0.6g e^{-kt} - g$$

$$v = \frac{0.6}{k} g e^{-kt} + \frac{g}{k}$$

$$v = \frac{g}{k} [0.6e^{-kt} + 1]$$

$$v = \frac{dx}{dt}, \text{ so } x = \int \frac{g}{k} [0.6e^{-kt} + 1] dt$$

$$x = \frac{g}{k} \left[-\frac{1}{k} 0.6e^{-kt} + t \right] + c$$

when $x = 0$ $t = 0$

$$0 = \frac{g}{k} \left[-\frac{1}{k} 0.6 \right] + c$$

$$c = \frac{g}{k^2} \times 0.6$$

$$\therefore x = \frac{g}{k} \left[-\frac{1}{k} 0.6e^{-kt} + t \right] + \frac{g}{k^2} (0.6)$$

when $x = D$, $v = 1.1w$ and $t = \frac{1}{k} \ln 6$

$$\begin{aligned} D &= \frac{g}{k} \left[-\frac{1}{k} 0.6 e^{-\ln 6} + \frac{1}{k} \ln 6 \right] + \frac{g}{k^2} (0.6) \\ &= \frac{-g}{k^2} 0.6 \times \frac{1}{6} + \frac{g}{k^2} \ln 6 + \frac{g}{k^2} (0.6) \\ &= \frac{g}{k^2} (0.6 - 0.1 + \ln 6) \\ &= \frac{g}{k^2} \left(\frac{1}{2} + \ln 6 \right) \end{aligned}$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Starting with one side, uses a double-angle result, or equivalent merit	1

Sample answer:

$$\cot x - \cot 2x$$

$$= \frac{\cos x}{\sin x} - \frac{\cos 2x}{\sin 2x}$$

$$= \frac{\cos x}{\sin x} - \frac{\cos^2 x - \sin^2 x}{2 \sin x \cos x}$$

$$= \frac{2 \cos^2 x - (\cos^2 x - \sin^2 x)}{2 \sin x \cos x}$$

$$= \frac{\cos^2 x + \sin^2 x}{2 \sin x \cos x} = \frac{1}{\sin 2x}$$

$$= \operatorname{cosec} 2x$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the fact that $2^{k+1} = 2 \times 2^k$ when establishing the inductive step, or equivalent merit	1

Sample answer:

Statement is true when $n = 1$ (from part (i)).

Assume statement is true for $n = k$.

That is $\operatorname{cosec}(2x) + \operatorname{cosec}(4x) + \cdots + \operatorname{cosec}(2^k x)$

$$= \cot x - \cot(2^k x)$$

Then

$$\operatorname{cosec}(2x) + \operatorname{cosec}(4x) + \cdots + \operatorname{cosec}(2^k x) + \operatorname{cosec}(2^{k+1} x)$$

$$= \cot x - \cot(2^k x) + \operatorname{cosec}(2^{k+1} x) \quad (\text{by assumption})$$

$$= \cot x - \cot(2^k x) + \cot(2^k x) - \cot(2^{k+1} x)$$

Using part (i)

$$\text{since } 2^{k+1}x = 2 \times 2^k x$$

$$= \cot x - \cot(2^{k+1} x) \quad \text{as required.}$$

Therefore statement is true for all $n \geq 1$ by induction.

Question 15 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Uses the substitution $u = -x$, or equivalent merit	1

$$\text{Let } I = \int_{-a}^a \frac{f(x)}{f(x) + f(-x)} dx$$

Let $u = x$ so that $du = -dx$.

$$\begin{aligned} I &= \int_a^{-a} \frac{f(-u)}{f(-u) + f(u)} (-du) \\ &= \int_{-a}^a \frac{f(-u)}{f(u) + f(-u)} du = \int_{-a}^a \frac{f(-x)}{f(x) + f(-x)} dx \end{aligned}$$

Question 15 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses part (i), or equivalent merit	1

Sample answer:

From part (i).

$$\begin{aligned} \int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx &= \int_{-1}^1 \frac{e^{-x}}{e^{-x} + e^x} dx \\ \therefore 2 \int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx &= \int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx + \int_{-1}^1 \frac{e^{-x}}{e^x + e^{-x}} dx \\ &= \int_{-1}^1 \frac{e^x + e^{-x}}{e^x + e^{-x}} dx \\ &= \int_{-1}^1 dx \\ &= [x]_{-1}^1 = 2 \\ \therefore \int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx &= 1 \end{aligned}$$

Question 15 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

The probability that A wins on the first draw is $\frac{1}{2}$. If A doesn't win on the first draw then A still has a further chance of winning.

$$\therefore P(A \text{ wins}) > \frac{1}{2}$$

$$\therefore P(B \text{ wins}) < \frac{1}{2}$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds the limiting sum for $P(A)$ or $P(B)$ and forms an inequality, or equivalent merit	2
• Finds an expression for the probability that one player wins, or equivalent merit	1

Sample answer:

$$P(B) = \frac{y}{w+y} \times \frac{y}{w+y} + \frac{y}{w+y} \times \frac{w}{w+y} \times \frac{y}{w+y} \times \frac{y}{w+y} + \dots$$

$$\begin{aligned} \text{Limiting sum} &= \frac{\frac{y^2}{(w+y)^2}}{1 - \frac{wy}{(w+y)^2}} \\ &= \frac{y^2}{(w+y)^2 - wy} \end{aligned}$$

For B to have a greater chance,

$$P(B) > \frac{1}{2}$$

$$\therefore \frac{y^2}{(w+y)^2 - wy} > \frac{1}{2}$$

$$2y^2 > w^2 + 2wy + y^2 - wy$$

$$y^2 - wy - w^2 > 0$$

$$\left(\frac{y}{w}\right)^2 - \frac{y}{w} - 1 > 0$$

$$\text{Roots of quadratic are } \frac{1 \pm \sqrt{5}}{2}$$

$$\frac{y}{w} > \frac{1 + \sqrt{5}}{2} \quad (\text{Positive root of quadratic}).$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Finds correct primitive, or equivalent merit	1

Sample answer:

Let $u = x + 1$

$$du = dx$$

$$\int_0^1 \frac{x}{(x+1)^2} dx$$

$$= \int_1^2 \frac{u-1}{u^2} du$$

$$= \int_1^2 u^{-1} - u^{-2} du$$

$$= [\ln u + u^{-1}]_1^2$$

$$= (\ln 2 + 2^{-1}) - (\ln 1 + 1)$$

$$= \ln 2 + \frac{1}{2} - 1$$

$$= \ln 2 - \frac{1}{2}$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Completes integration by parts and writes the result in terms of I_n, I_{n-1} and constants, or equivalent merit	2
• Attempts to use integration by parts, or equivalent merit	1

Sample answer:

$$I_n = \int_0^1 \frac{x^n}{(x-1)^2} dx$$

$$= \int_0^1 x^n \cdot \frac{d}{dx} \left(\frac{-1}{x+1} \right) dx \quad (\text{Integrating by parts})$$

$$= \left[\frac{-x^n}{x+1} \right]_0^1 + n \int_0^1 \frac{x^{n-1}}{x+1} dx$$

$$= \frac{-1}{2} + n \int_0^1 \frac{x^{n-1}}{(x+1)^2} \cdot (x+1) dx$$

$$= \frac{-1}{2} + n \int_0^1 \frac{x^n}{(x+1)^2} + \frac{x^{n-1}}{(x+1)^2} dx$$

$$\therefore I_n = -\frac{1}{2} + nI_n + nI_{n-1}$$

$$\therefore I_n = \frac{1}{2(n-1)} - \frac{n}{n-1} I_{n-1}$$

Question 15 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Finds I_2 , or equivalent merit	1

Sample answer:

$$I_1 = \ln 2 - \frac{1}{2}$$

$$I_2 = \frac{1}{2} - 2\left(\ln 2 - \frac{1}{2}\right)$$

$$= \frac{3}{2} - 2\ln 2$$

$$I_3 = \frac{1}{4} - \frac{3}{2}\left(\frac{3}{2} - 2\ln 2\right)$$
$$= 3\ln 2 - 2$$

Question 16 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains a cubic in $\cos \theta$, p , q , or equivalent merit	1

Sample answer:

Let $x = r \cos \theta$

$$r = \sqrt{\frac{4p}{3}} \quad \text{and so} \quad r^3 = \frac{4p}{3}r$$

$$x^3 - px + q = r^3 \cos^3 \theta - pr \cos \theta + q$$

$$= \frac{4p}{3}r \cos^3 \theta - pr \cos \theta + q$$

$$= \frac{pr}{3}(4 \cos^3 \theta - 3 \cos \theta) + q$$

$$= \frac{pr}{3} \cos 3\theta + q$$

$$= \frac{pr}{3} \left(\frac{-4q}{r^2} \right) + q$$

$$= \frac{-4pq}{3} \frac{3}{4p} + q$$

$$= 0$$

So $r \cos \theta$ is a root of $x^3 - px + q = 0$.

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Writes down a cubic with roots $\alpha + 3$, $\beta + 3$, $\gamma + 3$, or equivalent merit	1

Sample answer:

$\alpha + 3$, $\beta + 3$, $\gamma + 3$ will be roots of

$$0 = (x-3)^3 + 9(x-3)^2 + 15(x-3) - 17$$

$$= x^3 - 9x^2 + 27x - 27 + 9x^2 - 54x + 81 + 15x - 45 - 17$$

$$= x^3 - 12x - 8$$

Question 16 (a) (iii)

Criteria	Marks
• Provides correct solution	3
• Finds three values for θ , or equivalent merit	2
• Calculates r (from part (i)), or equivalent merit	1

Sample answer:

Use $p = 12$, $q = -8$ in (i)

$$r = \sqrt{\frac{4p}{3}} = \sqrt{16} = 4$$

$$\cos 3\theta = \frac{-4(-8)}{4^3} = \frac{1}{2}$$

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$

$$\text{So } 4\cos\frac{\pi}{9}, 4\cos\frac{5\pi}{9} \text{ and } 4\cos\frac{7\pi}{9}$$

are roots of $x^3 - 12x - 8 = 0$,

and so the roots of

$$x^3 + 9x^2 + 15x - 17 = 0$$

$$\text{are } 4\cos\frac{\pi}{9} - 3, 4\cos\frac{5\pi}{9} - 3, 4\cos\frac{7\pi}{9} - 3$$

Question 16 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Simplifies the sum of the products of pairs of roots, or equivalent merit	2
• Finds the sum of the roots, or equivalent merit	1

Sample answer:

$$\alpha + \bar{\alpha} + \beta + \bar{\beta} = 2k$$

$$2\operatorname{Re}(\alpha) + 2\operatorname{Re}(\beta) = 2k \quad \text{①}$$

$$\alpha\bar{\alpha} + \alpha\beta + \alpha\bar{\beta} + \bar{\alpha}\beta + \bar{\alpha}\bar{\beta} + \beta\bar{\beta} = 2k^2$$

$$\alpha(\beta + \bar{\beta}) + \bar{\alpha}(\beta + \bar{\beta}) = 2k^2 - 2$$

$$2\alpha\operatorname{Re}(\beta) + 2\bar{\alpha}\operatorname{Re}(\beta) = 2k^2 - 2$$

$$2\operatorname{Re}(\beta)(\alpha + \bar{\alpha}) = 2k^2 - 2$$

$$4\operatorname{Re}(\alpha)\operatorname{Re}(\beta) = 2k^2 - 2$$

$$2\operatorname{Re}(\alpha)\operatorname{Re}(\beta) = k^2 - 1$$

From ①

$$2\operatorname{Re}(\alpha) + 2\operatorname{Re}(\beta) = 2k$$

$$\operatorname{Re}(\alpha) + \operatorname{Re}(\beta) = k$$

$$\operatorname{Re}(\alpha)^2 + 2\operatorname{Re}(\alpha)\operatorname{Re}(\beta) + \operatorname{Re}(\beta)^2 = k^2$$

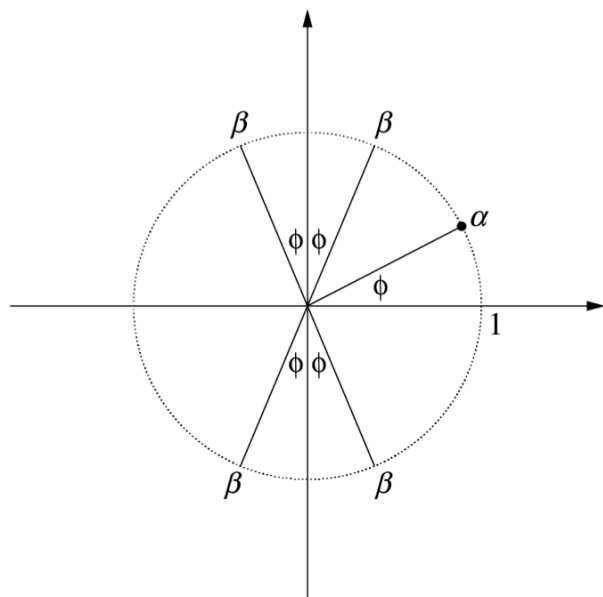
$$\operatorname{Re}(\alpha)^2 + k^2 - 1 + \operatorname{Re}(\beta)^2 = k^2$$

$$\text{So } \operatorname{Re}(\alpha)^2 + \operatorname{Re}(\beta)^2 = 1$$

Question 16 (b) (ii)

Criteria	Marks
• Provides correct sketch	2
• Finds one correct possibility for β , or equivalent merit	1

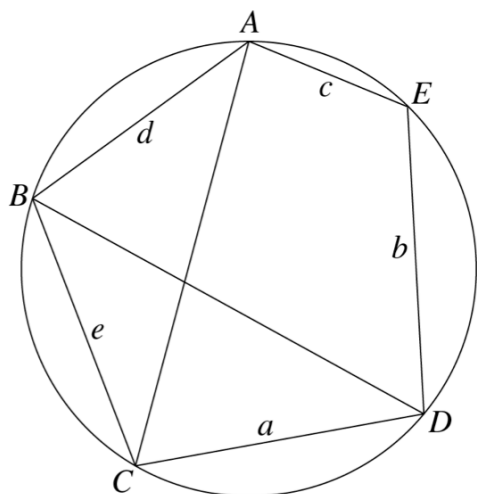
Sample answer:



Question 16 (c)

Criteria	Marks
• Provides correct solution	3
• Shows that $\frac{a}{\sin(B+E-\pi)} = \frac{d}{\sin(C+E-\pi)}$, or equivalent merit	2
• Shows that $\frac{a}{\sin DBC} = \frac{d}{\sin ACB}$, or equivalent merit	1

Sample answer:



Sine rule in $\triangle ABC$

$$\frac{d}{\sin ACB} = \frac{e}{\sin BAC}$$

in $\triangle DBC$

$$\frac{a}{\sin DBC} = \frac{e}{\sin BDC}$$

But $\angle BDC$ and $\angle BAC$ stand on same chord and so are equal.

$$\frac{a}{\sin DBC} = \frac{e}{\sin BDC} = \frac{e}{\sin BAC} = \frac{d}{\sin ACB}$$

$ABDE$ is cyclic so

$$E + ABD = \pi$$

$$\text{but } B = ABD + DBC$$

$$= \pi - E + DBC$$

$$DBC = B + E - \pi$$

$$\sin DBC = \sin(B + E - \pi)$$

$$= -\sin(B + E)$$

similarly $ACB = (C + E - \pi)$ ($C - ACD$, and $ACD = ABD = \pi - E$)

$$\sin ACB = -\sin(C + E)$$

$$\frac{a}{\sin DBC} = \frac{d}{\sin ACB}$$

$$-\frac{a}{\sin(B + E)} = -\frac{d}{\sin(C + E)}$$

$$\text{Hence } \frac{a}{\sin(B + E)} = \frac{d}{\sin(C + E)}$$

2019 HSC Mathematics Extension 2 Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	2.1	E3
2	1	4.1	E8
3	1	4.1	E8
4	1	7.2	E4
5	1	2.5, 3.1	E3
6	1	1.7	E6
7	1	1.2, 1.6, 4.1	E6
8	1	2.2	E3
9	1	1.5, 8	E6
10	1	8	E2

Section II

Question	Marks	Content	Syllabus outcomes
11 (a) (i)	1	2.1	E3
11 (a) (ii)	2	2.1	E3
11 (b)	3	3.1	E3
11 (c)	2	4.1	E8
11 (d)	3	4.1	E8
11 (e) (i)	2	2.2	E3
11 (e) (ii)	2	2.2, 2.4	E3
12 (a)	2	2.5	E3
12 (b) (i)	1	8	E2
12 (b) (ii)	3	1.8, 8	E6
12 (c)	4	4.1, 5.1	E7, E8
12 (d) (i)	1	1.3	E6
12 (d) (ii)	2	1.5	E6
12 (d) (iii)	2	1.5	E6
13 (a) (i)	2	3.2	E3, E4
13 (a) (ii)	2	3.2	E4
13 (a) (iii)	3	3.2	E4
13 (b) (i)	2	5.1	E8
13 (b) (ii)	1	5.1	E8
13 (c) (i)	2	6.1	E5
13 (c) (ii)	3	6.1	E5
14 (a)	3	6.3.4	E5
14 (b) (i)	1	6.2.3	E5

Question	Marks	Content	Syllabus outcomes
14 (b) (ii)	4	6.2.3	E5
14 (b) (iii)	3	6.2.3	E5
14 (c) (i)	2	8	E2, E9
14 (c) (ii)	2	8.2	E2, E9
15 (a) (i)	2	4.1	E8
15 (a) (ii)	2	4.1	E8
15 (b) (i)	1	8	E9
15 (b) (ii)	3	8	E9
15 (c) (i)	2	4.1	E8
15 (c) (ii)	3	4.1	E8
15 (c) (iii)	2	4.1	E8
16 (a) (i)	2	7.5	E4
16 (a) (ii)	2	7.5	E4
16 (a) (iii)	3	7.5	E4
16 (b) (i)	3	2	E3, E4
16 (b) (ii)	2	2	E3
16 (c)	3	8.1	E9